

PSet 5 Writeup

Final Part 2 writeup only, excluding code cells.

Part 2

Filtered to the README submission checklist only. Includes written responses and required figure files when present.

Problem 1.1 and 1.2

Suggested Derivations

1.

$$q(x_t | x_0) = \mathcal{N}(x_t; \sqrt{\bar{\alpha}_t} x_0, (1 - \bar{\alpha}_t) \mathbf{I})$$

Brief proof: write $x_t = \sqrt{\alpha_t} x_{t-1} + \sqrt{\beta_t} \epsilon_t$ and expand recursively. The coefficient on x_0 becomes $\prod_{i=1}^t \sqrt{\alpha_i} = \sqrt{\bar{\alpha}_t}$, while the accumulated Gaussian variance becomes $1 - \bar{\alpha}_t$.

2.

$$\lim_{T \rightarrow \infty} q(x_T) = \mathcal{N}(0, \mathbf{I})$$

Under any normal schedule, infinitely many β_t stay bounded away from 0, so $\bar{\alpha}_T = \prod_{i=1}^T (1 - \beta_i) \rightarrow 0$. From part (1), the mean goes to 0 and the covariance goes to $\mathbf{I}$.

3.

$$\tilde{\beta}_t = \frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t} \beta_t$$

4. Writing $\tilde{\mu}_t(x_t, x_0) = A x_t + B x_0$,

$$A = \frac{\sqrt{\alpha_t} (1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t}, \quad B = \frac{\sqrt{\bar{\alpha}_{t-1}} \beta_t}{1 - \bar{\alpha}_t}$$

5.

$$x_0 = \frac{x_t - \sqrt{1 - \bar{\alpha}_t} \epsilon_t}{\sqrt{\bar{\alpha}_t}}$$

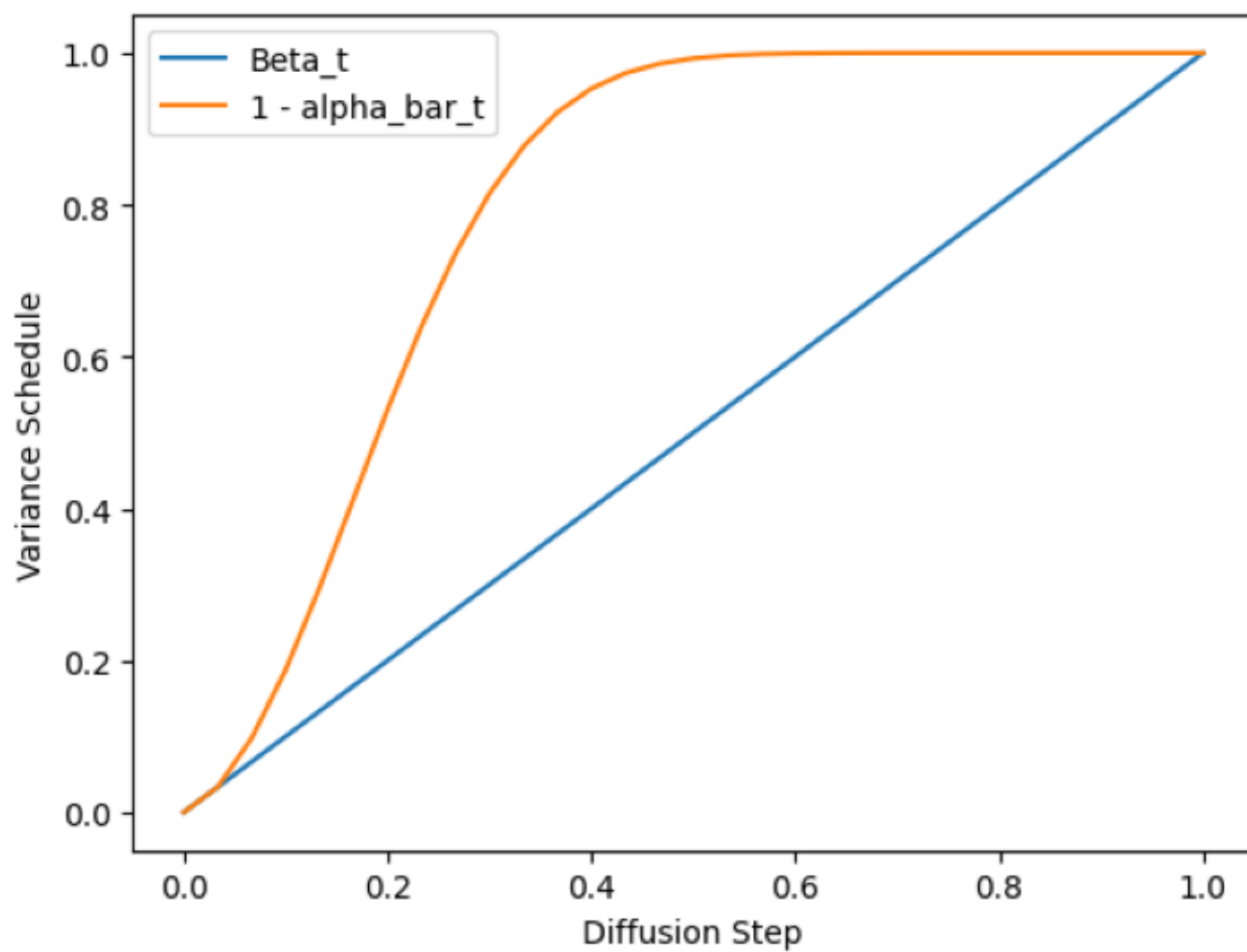
6. Substituting (5) into (4),

$$\tilde{\mu}_t(x_t, \epsilon_t) = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_t \right)$$

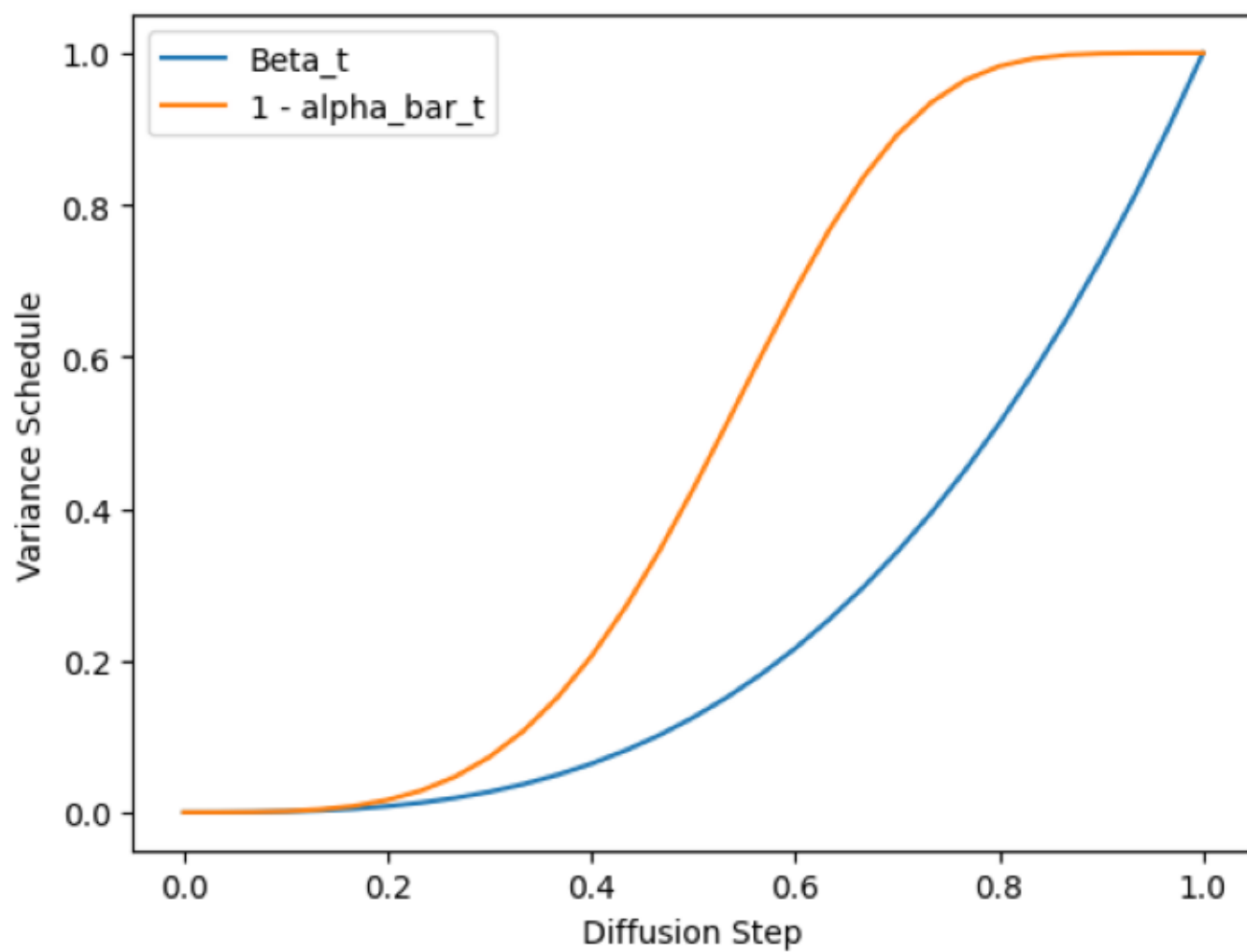
So

$$A' = \frac{1}{\sqrt{\alpha_t}}, \quad B' = -\frac{\beta_t}{\sqrt{\alpha_t} \sqrt{1 - \bar{\alpha}_t}}$$

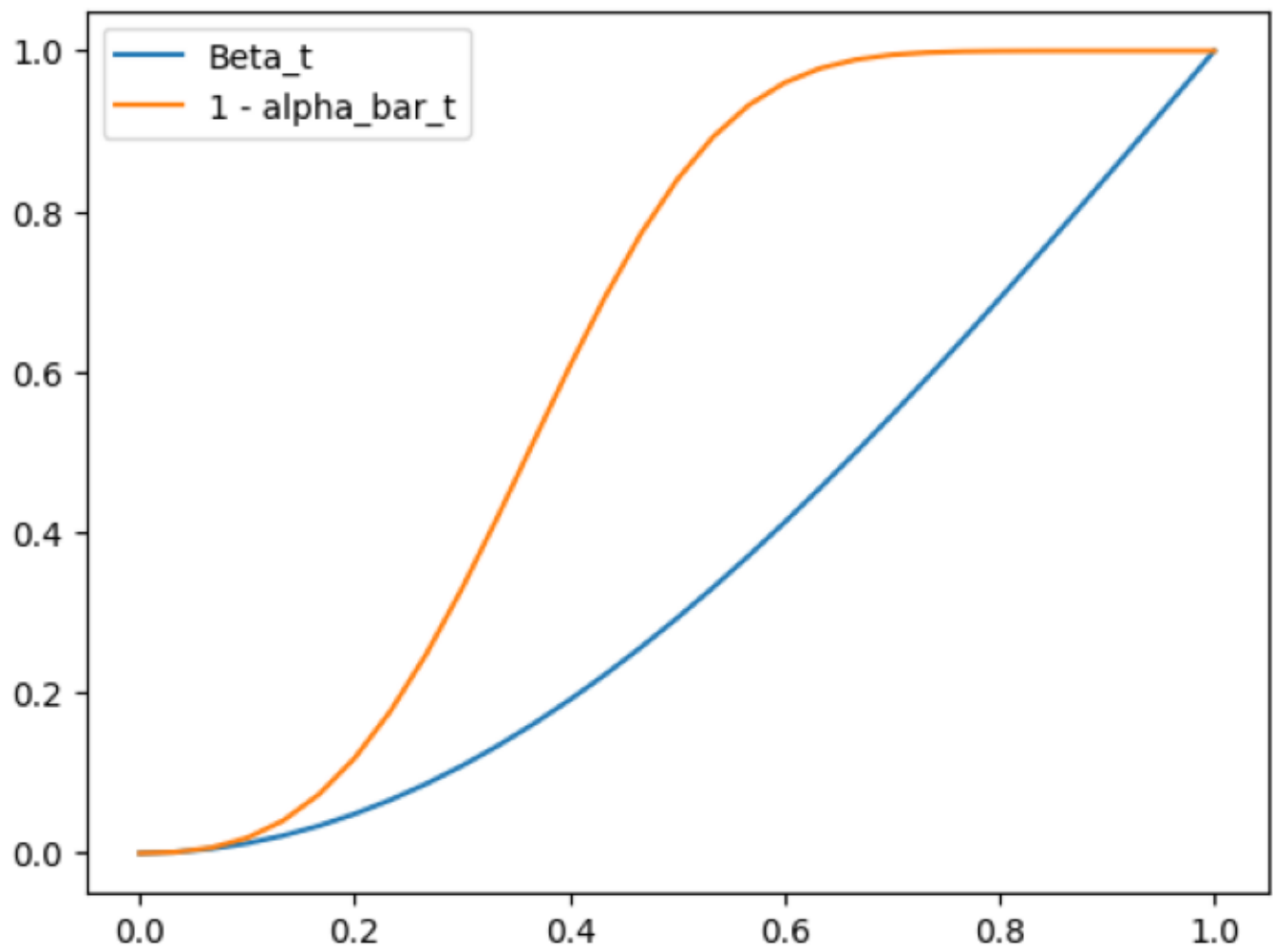
Problem 1 Figure 1



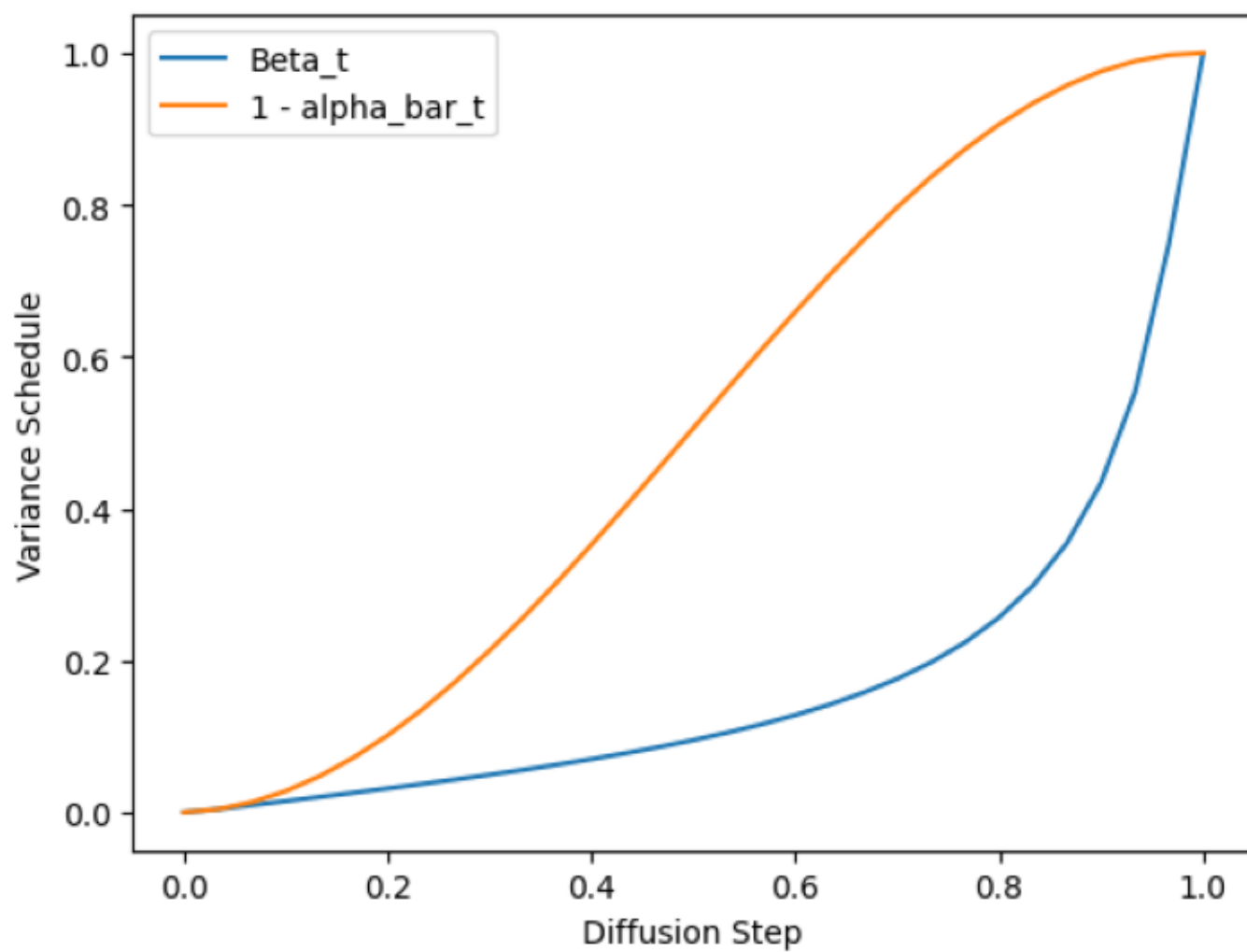
Problem 1 Figure 2



Problem 1 Figure 3



Problem 1 Figure 4

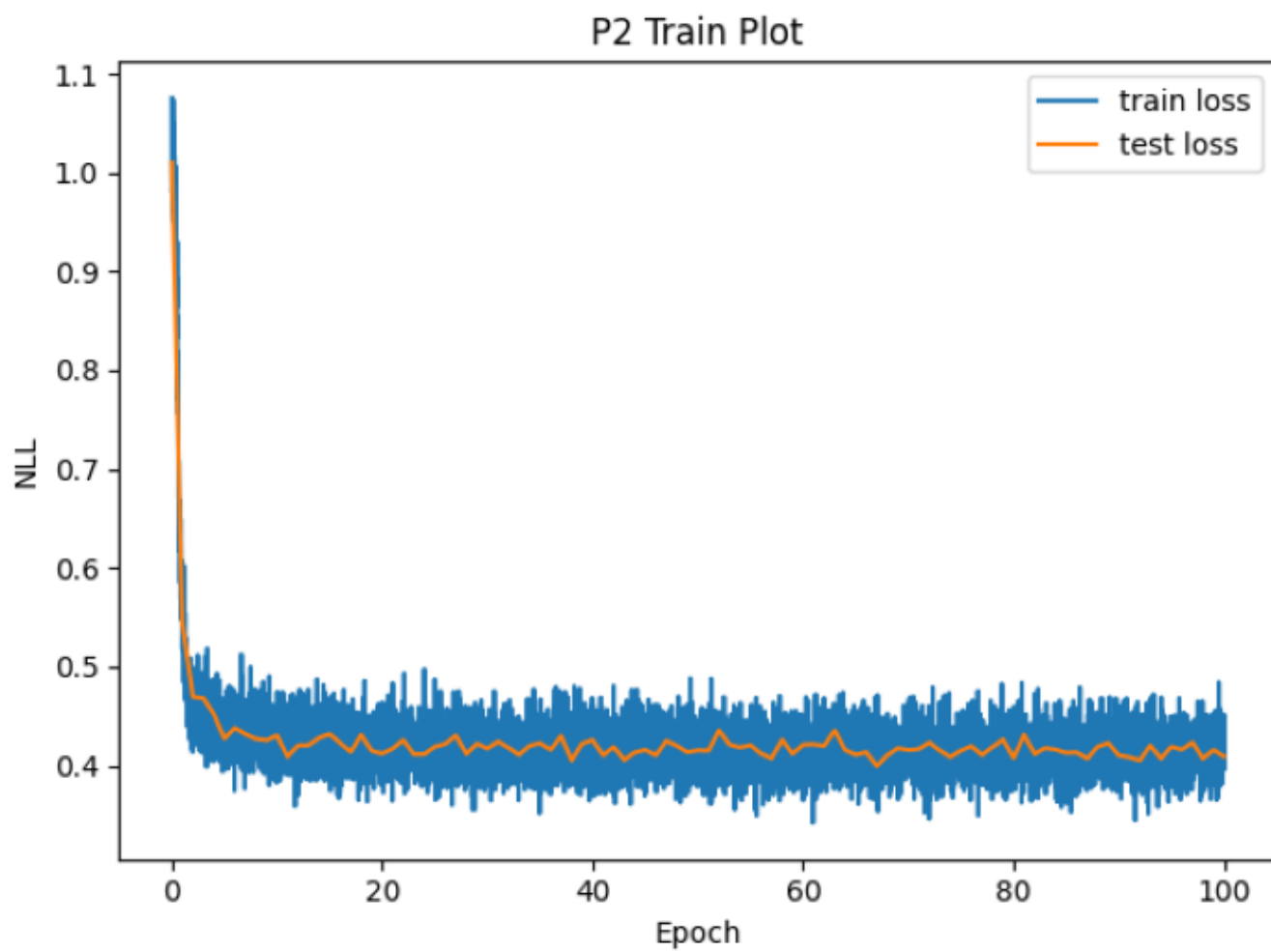


Problem 1 Schedule Discussion

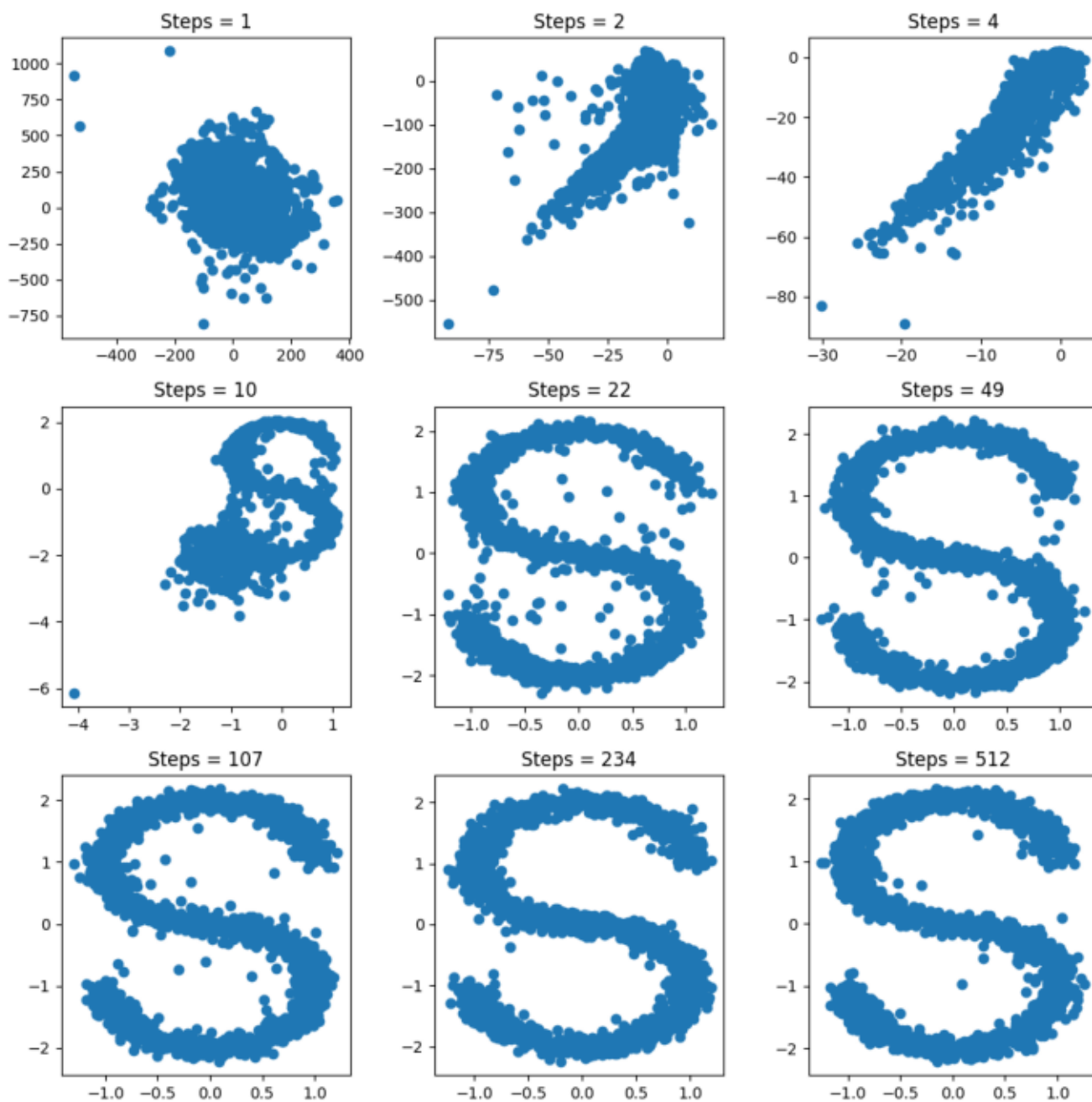
****Schedule Discussion****

Compared with the linear and polynomial schedules, the Nichol-Dhariwal cosine schedule changes cumulative noise level more smoothly. It avoids corrupting the data too aggressively at early timesteps, keeps useful signal around for more timesteps, and also avoids an abrupt collapse near the end of the chain. That gives the denoiser a better spread of training examples across easy, medium, and hard noise levels, which usually leads to more stable optimization and better final samples.

Problem 2: p2_train_plot.png



Problem 2: p2_toy_samples.png

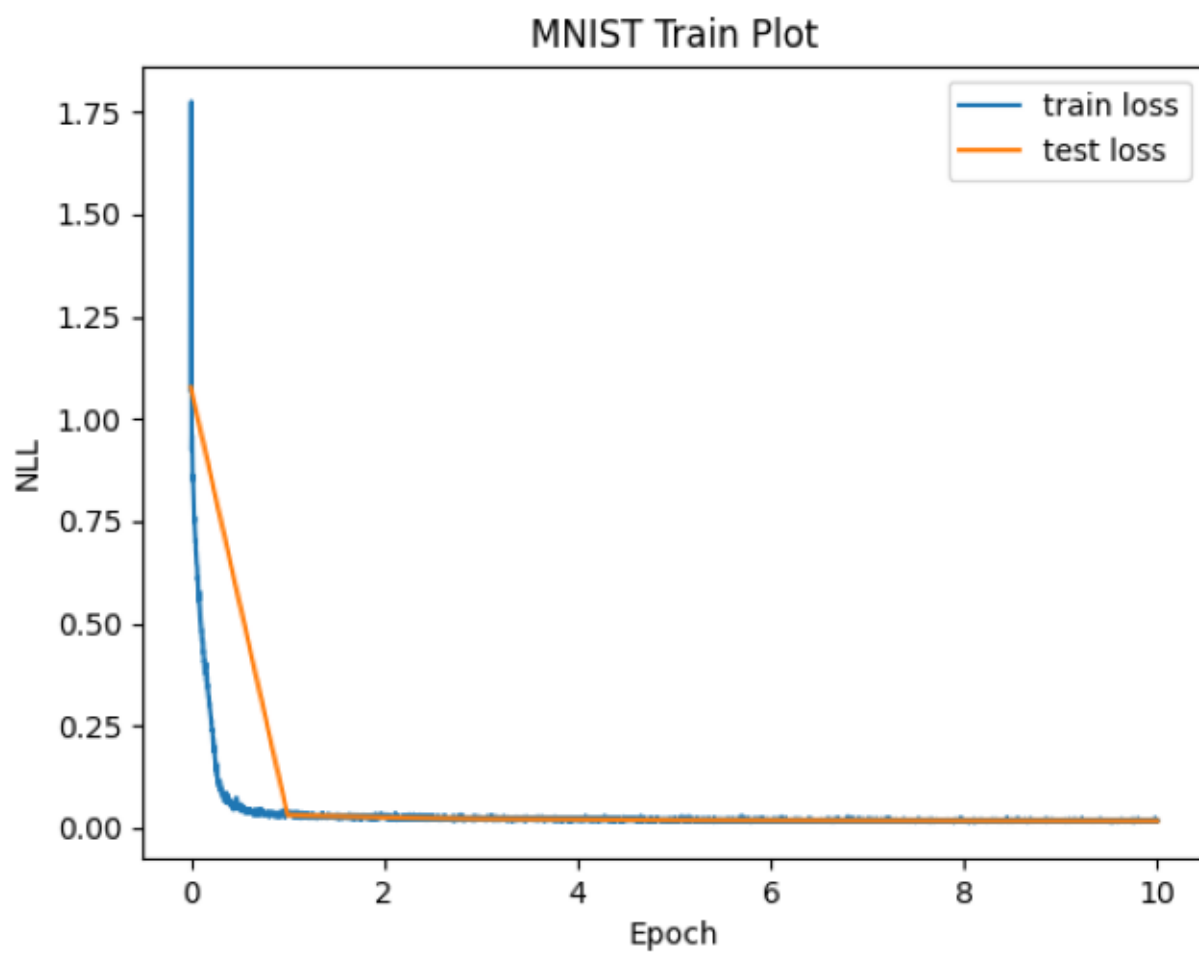


Problem 3 U-Net Architecture

****Architecture Explanation****

- ``FourierEncoder``: maps the scalar timestep into a higher-dimensional periodic embedding so the network can condition every layer on where the sample is in the diffusion process. Input is a batch of timesteps; output is a batch of sinusoidal feature vectors.
- ``ResidualLayer``: refines a feature map while injecting both the time embedding and the label embedding. It takes an image feature tensor plus two embedding vectors, runs two conv-norm-activation blocks, adds the broadcasted embeddings, and then adds the residual connection back.
- ``Encoder``: keeps the same channel count through several residual layers and then downsamples to a coarser spatial resolution with more channels. Its role is to compress the image into increasingly abstract semantic features.
- ``Midcoder``: processes the bottleneck representation at the lowest resolution without changing the spatial size. It lets the network mix global information before decoding.
- ``Decoder``: upsamples the coarse representation back to a finer resolution and refines it with several residual layers conditioned on time and label. Its role is to reconstruct detailed spatial structure while using the information gathered by the encoder path.

Problem 3: mnist_train_plot.png



Problem 3: image_w0.0.png



Problem 3: image_w0.5.png

0	1	2	3	4	5	6	7	8	9
0	1	2	3	4	5	6	7	8	9
0	1	2	3	4	5	6	7	8	9
0	1	2	3	4	5	6	7	8	9

Problem 3: image_w1.0.png

0	1	2	3	4	5	6	7	8	9
0	1	2	3	4	5	6	7	8	9
0	1	2	3	4	5	6	7	8	9
0	1	2	3	4	5	6	7	8	9

Problem 3: image_w2.0.png

0	1	2	3	4	5	6	7	8	9
0	1	2	3	4	5	6	7	8	9
0	1	2	3	4	5	6	7	8	9
0	1	2	3	4	5	6	7	8	9

Problem 3: image_w4.0.png

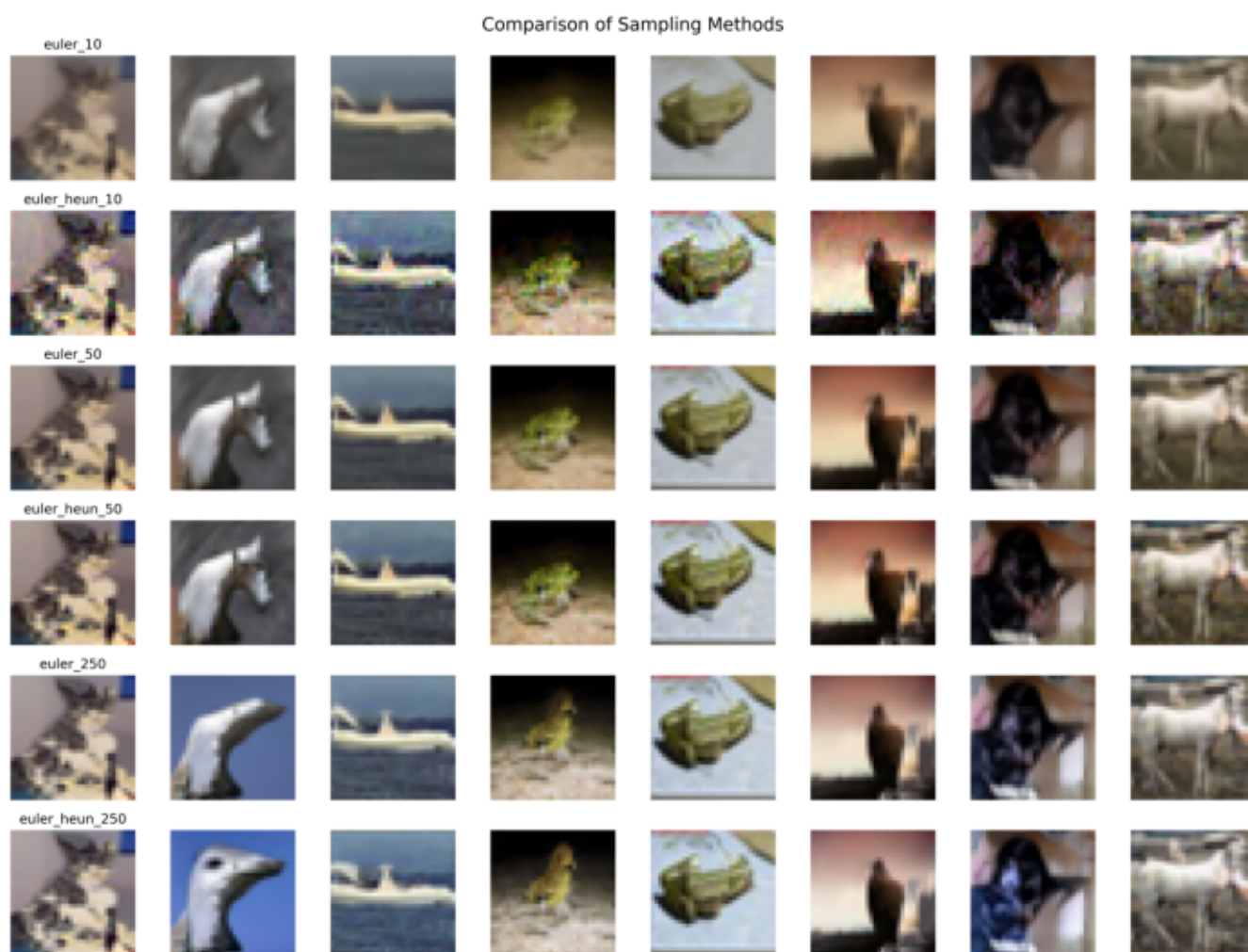
0	1	2	3	4	5	6	7	8	9
0	1	2	3	4	5	6	7	8	9
0	1	2	3	4	5	6	7	8	9
0	1	2	3	4	5	6	7	8	9

Problem 3 CFG Discussion

****CFG Weight Discussion****

Small guidance weights keep the samples diverse but the class identity can be weaker. Moderate guidance usually gives the best tradeoff: digits become sharper and better aligned with the requested label without collapsing too much. Very large guidance often over-constrains the reverse process, which can reduce diversity and introduce artifacts or unnatural-looking digits.

Problem 4: sampling_comparison.png



Problem 4 Sampler Discussion

****Conclusion****

At a fixed step budget, more denoising steps usually improve sample fidelity because the ODE is integrated more accurately. Heun correction typically helps most when the step count is small: it reduces the local truncation error of plain Euler, so the samples are usually cleaner and more stable at 10 or 50 steps. As the number of steps becomes large, the gap narrows because both samplers are already close to the continuous trajectory.